

Executive Summary

Marvel MCU SQL Business Analysis

Project Overview

The **Marvel MCU SQL Business Analysis** project is an end-to-end SQL portfolio project focused on transforming a raw Marvel content dataset into structured, business-ready insights. The project analyzes Marvel's content catalog across multiple dimensions including **content volume, release trends, Marvel universes, MCU canon status, content type, animation status, ratings, and cast presence**.

The analysis was designed as a progressive business-analysis journey, moving from foundational descriptive analysis to comparative, trend-based, performance-oriented, and cross-dimensional insights. The project ultimately demonstrates how SQL can be used not only to retrieve data, but also to answer practical business questions and support data-driven decision-making.

Business Objective

The primary objective of the project was to understand the **scale, composition, performance, and evolution of Marvel's content portfolio** and identify patterns that could help evaluate content strategy and performance.

The analysis focused on questions such as:

1. How large is Marvel's content catalog and how is it distributed across movies, TV series, universes, decades, and animation categories?
2. How has Marvel's content production evolved over time?
3. How does content performance differ across Marvel universes and content types?
4. How do Marvel titles perform across different rating platforms?
5. Which universes, content categories, and release periods demonstrate stronger rating performance?
6. Which content segments combine high production volume with strong ratings?
7. Which actors have the strongest recurring presence across Marvel's content ecosystem?
8. Which actors have appeared across multiple Marvel universes?
9. Which Marvel universes demonstrate the largest and most diverse cast presence?

Data Preparation & Structure

Before beginning Business Analysis, the project underwent dedicated **data auditing and cleaning**, resulting in three validated analytical tables:

- `marvel_master_clean` — Core Marvel title, release, universe, content type, canon, and animation information.
- `marvel_ratings_clean` — Ratings collected across multiple rating sources.
- `marvel_cast_clean` — Cast and actor-level information associated with Marvel titles.

These tables provide separate but connected analytical perspectives, enabling both single-table analysis and multi-table SQL analysis.

Analytical Approach

The Business Analysis was structured into three progressive stages.

Basic Business Analysis

The Basic stage established a foundation through:

- Catalog size and composition
- Movies vs TV Series distribution
- Content by year and decade
- Marvel universe distribution
- MCU vs non-MCU content
- Animated vs non-animated content
- Rating coverage
- Average, highest, and lowest ratings

This stage established the overall structure and characteristics of the Marvel content portfolio.

Intermediate Business Analysis

The Intermediate stage moved beyond simple counts into:

- Content production trends
- Universe-level performance
- Movie vs TV Series comparisons
- Animated vs non-animated rating comparisons
- MCU vs non-MCU performance
- Rating coverage comparisons

- Release-year rating trends
- Older vs newer content performance

This stage focused on identifying **relationships, comparisons, trends, and performance differences** within the Marvel portfolio.

Advanced Business Analysis

The Advanced stage was intentionally designed to remain **beginner-friendly while demonstrating more sophisticated business thinking.**

The analysis covered:

- Highest-rated Marvel titles
- Cross-source rating differences
- Titles performing above the overall Marvel rating benchmark
- Universe-level rating performance
- Content volume vs rating performance
- High-volume but below-average-performing universes
- Movie vs TV Series performance
- Animated vs non-animated performance
- Release-period performance
- Universe + content-type combinations
- Actor participation and recurring presence
- Cross-universe actor presence
- Cast associated with highly rated content
- Universe-level cast diversity

This extended the project from content and ratings analysis into **portfolio performance and talent analysis.**

Rating Normalization

Because the rating sources use different scoring scales, cross-source comparisons were standardized before calculating averages or differences.

The project applied a common **10-point rating scale**:

- **IMDb** → unchanged
- **TMDB** → unchanged
- **Rotten Tomatoes** → divided by 10

- **Metacritic** → divided by 10

This ensured that cross-source rating comparisons were analytically meaningful and prevented differences in scoring scales from distorting the results.

Key Insights

01- MCU Dominates Marvel's Content Portfolio

The MCU accounts for 54.7% of the entire Marvel content catalog, making it the largest universe by a significant margin.

Source: Q1, Q6

02- Marvel's Content Production Exploded in the 2010s

Marvel's content output increased by 222.7% from the 2000s to the 2010s, reaching 71 titles—the highest volume of any decade.

Source: Q11, Q21

03- 2018 Was Marvel's Peak Production Year

2018 recorded the highest annual content output with 13 titles, followed by 2017 with 12.

Source: Q10, Q22

04- MCU Combines Scale with Strong Performance

The MCU has both the largest content portfolio (88 titles) and the highest average normalized rating (6.99) among Marvel universes.

Source: Q42, Q43

05- The 2010s Were Marvel's Strongest Era

The 2010s delivered Marvel's strongest combination of scale and reception: 71 titles with an average normalized rating of 7.12.

Source: Q47

06- Newer Marvel Content Consistently Outperforms Older Content

Newer content achieved higher average ratings across every major platform, with Rotten Tomatoes showing the largest improvement—from 62.53 to 71.97.

Source: Q38

07- Rating Perception Can Vary Dramatically Across Platforms

Some Marvel titles show rating gaps approaching 5 points between their highest and lowest normalized scores, highlighting significant differences in platform perception.

Source: Q40

08- Spider-Verse Leads Marvel's Highest-Rated Content

Spider-Man: Into the Spider-Verse and Spider-Man: Across the Spider-Verse occupy the top two positions among the highest-rated Marvel titles, with normalized ratings of 8.80 and

8.74.

Source: Q39

09- Netflix Marvel Has a Distinctly TV-Driven Strategy

Netflix Marvel stands apart from other universes, with 12 TV series versus just 1 movie—a 0.08 movie-to-TV ratio.

Source: Q24

10- High Content Volume Does Not Guarantee Strong Ratings

Pre-MCU and Netflix Marvel fall below the overall normalized rating benchmark despite having substantial content portfolios.

Source: Q44

11- A Small Group of Actors Connects Multiple Marvel Universes

Nicolas Cage and Oscar Isaac have appeared across three Marvel universes, demonstrating how select talent can bridge otherwise distinct Marvel properties.

Source: Q50

12- MCU Has the Broadest Cast Ecosystem

The MCU has 552 unique actors in the analyzed cast data—far more than any other Marvel universe.

Source: Q52

13- Chris Evans Has the Strongest Recurring Marvel Presence

Chris Evans appears in the highest number of distinct Marvel titles, with 10 appearances, followed by Robert Downey Jr. and Samuel L. Jackson with 9 each.

Source: Q49

14 - MCU Movies Outperform MCU Series on Ratings

Within the MCU, movies achieve a higher average normalized rating (7.01) than TV series (6.78).

Source: Q48

SQL Skills Demonstrated

The project demonstrates practical SQL skills including:

Data Aggregation

COUNT() • AVG() • MIN() • MAX() • COUNT(DISTINCT)

Data Transformation

CASE • TRY_CAST() • Rating Normalization

Multi-Table Analysis

INNER JOIN • CROSS JOIN

Advanced SQL

CTEs • Window Functions • ROW_NUMBER() • LAG() • LEAD()

Business Analysis

Benchmark Analysis • Trend Analysis • Segmentation • Comparative Analysis

Performance Analysis

Ranking • Volume vs Rating • Period Comparison • Cross-Source Analysis

Recommendations

- **Balance Content Expansion with Quality** — Monitor whether increased production volume is accompanied by strong audience and critical reception.
- **Prioritize High-Performing Content Segments** — Identify successful universes, content types, and release periods that can inform future content strategy.
- **Investigate High-Volume, Low-Performance Segments** — Review areas where substantial content investment is not translating into strong ratings.
- **Use Cross-Platform Ratings Holistically** — Consider multiple rating sources rather than relying on a single platform when evaluating content performance.
- **Leverage Proven Talent** — Analyze recurring and cross-universe talent to identify opportunities for successful actor–franchise combinations.
- **Track Portfolio Performance Continuously** — Monitor content volume, ratings, universe performance, and cast trends as the Marvel portfolio evolves.

Project Outcome

From Raw Marvel Data to Business Insights

- Built an end-to-end SQL analysis using 3 cleaned and validated datasets.
- Answered 52 structured business questions across Basic, Intermediate, and Advanced analysis.
- Evaluated Marvel's content portfolio, release trends, universes, ratings, content types, animation, and cast ecosystem.
- Applied rating normalization to enable meaningful cross-platform comparisons.
- Progressed from descriptive analysis to comparative, trend-based, and multi-table business analysis.
- Transformed SQL query results into actionable business insights and recommendations.